

KHOLMSK, Russian Federation

WMO: 321280

Lat: **47.050N**Lon: **142.050E**Elev: **144**StdP: **14.62**Time Zone: **11.00 (E11)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.7	7.5	-4.6	4.3	7.5	-1.5	5.1	9.1	27.5	18.3	25.3	19.7	5.5	90	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	74.2	68.6	72.3	67.3	70.7	65.9	70.0	73.0	68.4	71.3	66.9	69.9	8.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.9	106.7	72.0	67.3	100.9	70.6	65.7	95.4	69.2	34.3	73.2	32.9	71.5	31.7	69.9	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.3	17.9	DB	2.5	77.8	3.7	2.3	-0.1	79.5	-2.3	80.8	-4.4	82.2	-7.0	83.9	(4)
(5)			WB	1.2	71.3	3.4	2.0	-1.2	72.7	-3.2	73.9	-5.1	75.0	-7.5	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	18.2	19.2	27.3	37.9	46.8	54.5	62.1	65.7	60.2	48.2	33.9	22.7
	DBStd	17.51	6.39	6.73	6.21	5.60	5.80	4.97	4.53	4.25	4.99	6.14	7.71	6.40
	HDD50	4504	986	861	702	366	135	15	0	0	1	107	484	847
	HDD65	8668	1451	1281	1167	814	565	316	110	44	155	520	933	1312
	CDD50	1412	0	0	0	2	35	151	377	486	308	52	1	0
	CDD65	100	0	0	0	0	0	1	22	65	12	0	0	0
	CDH74	59	0	0	0	0	0	1	11	43	4	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	8.1	8.0	8.5	7.9	7.0	5.1	4.7	5.9	7.0	8.8	9.5	8.8
Precipitation	PrecAvg	31.3	2.1	1.4	1.1	1.8	2.6	2.1	3.6	3.9	3.9	3.2	2.8	2.8
	PrecMax	42.1	4.3	3.0	3.1	5.5	4.8	5.5	9.9	13.5	7.7	8.1	8.1	5.9
	PrecMin	22.8	0.7	0.2	0.0	0.4	1.1	0.1	0.4	0.8	0.9	0.9	0.6	0.7
	PrecStd	5.6	0.9	0.7	0.7	1.1	1.0	1.1	1.9	2.7	1.7	1.5	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	38.4	44.0	57.4	65.9	70.7	75.5	77.4	73.9	64.5	55.3	42.4
		MCWB	33.0	35.9	38.9	49.0	54.7	61.0	67.5	69.8	68.2	58.5	52.2	40.2
	2%	DB	31.9	35.0	40.2	52.5	61.8	66.3	72.3	75.1	71.3	61.7	51.3	37.9
		MCWB	29.1	32.6	36.4	44.8	52.7	58.7	66.4	69.6	65.4	56.2	48.1	35.2
	5%	DB	29.3	32.2	38.3	49.1	58.4	64.1	70.5	73.5	69.6	59.3	48.0	34.6
		MCWB	26.9	29.5	35.0	42.7	50.6	58.0	65.4	68.6	64.1	54.6	44.2	32.0
	10%	DB	27.2	29.2	36.1	46.2	55.7	62.0	68.7	72.0	67.7	57.1	45.2	31.9
		MCWB	24.9	26.6	33.0	41.1	49.4	57.0	64.6	67.5	62.5	52.4	41.6	29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.2	36.4	39.9	50.9	57.4	63.8	70.1	72.7	69.4	59.8	52.1	40.1
		MCDB	35.2	37.7	42.7	55.7	62.5	67.4	73.0	75.0	72.6	63.5	54.7	42.2
	2%	WB	29.5	33.0	37.2	46.2	54.0	61.1	68.1	71.1	67.0	57.3	48.2	35.6
		MCDB	31.5	34.9	39.5	51.1	59.3	64.4	70.6	73.6	70.5	60.7	50.8	37.5
	5%	WB	27.1	29.6	35.3	43.6	52.1	59.3	66.7	69.7	64.9	55.3	44.6	32.3
		MCDB	29.1	31.9	37.9	47.9	56.9	62.6	69.2	72.7	68.6	58.7	47.3	34.4
	10%	WB	25.0	26.7	33.4	41.4	50.4	57.7	65.4	68.3	63.2	53.1	41.6	29.4
		MCDB	27.1	29.1	36.1	45.5	54.6	60.8	68.0	71.1	67.1	56.6	45.1	31.6
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.7	7.8	8.7	9.5	8.7	8.0	8.1	9.2	9.0	7.7	6.8
		MCDBR	8.5	9.7	9.1	12.1	12.9	11.5	9.4	8.7	10.0	10.0	10.1	10.3
		MCWBR	8.0	9.4	8.3	8.5	7.9	7.4	6.2	6.4	7.8	9.1	9.6	10.3
	5% WB	MCDBR	8.5	9.7	9.0	11.3	11.0	9.0	7.9	7.2	9.0	9.3	10.4	10.5
		MCWBR	8.3	9.6	8.6	8.7	7.8	6.9	6.0	6.1	8.1	9.6	10.8	10.7
Clear-Sky Solar Irradiance	taub	0.322	0.365	0.419	0.497	0.523	0.500	0.464	0.440	0.397	0.404	0.378	0.341	
	taud	2.078	1.958	1.891	1.831	1.873	2.033	2.212	2.289	2.339	2.145	2.097	2.043	
	Ebn at Noon	228	241	247	240	239	246	254	256	257	228	204	201	
	Edh at Noon	30	42	53	62	61	53	43	39	34	35	30	28	
All-Sky Solar Radiation	RadAvg	377	660	1074	1433	1536	1514	1414	1341	1211	801	436	296	
	RadStd	43	63	93	153	142	169	173	121	116	52	40	28	